Compartmental TB Diagnostics: A Simulated BAL Cytokine Cohort

Project background, simulation methodology, and data dictionary

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Note on the dataset: All patient records in the accompanying CSV are entirely simulated. No real patient data, real assay measurements, or identifiable information were used at any stage. Concentrations, correlation structure, and group effects were generated computationally to be broadly consistent with published patterns in the TB immunology literature (cited below), not copied or fitted from any specific dataset. This project is intended purely as a methods/skills demonstration.

1. Scientific Background

Most host-biomarker TB research is done on peripheral blood (e.g. QuantiFERON/IGRA-stimulated plasma), which is convenient to sample but reflects systemic, not local, immunity. Bronchoalveolar lavage (BAL) samples the lung compartment directly and shows a distinct, more compartmentalized immune signature. Studies comparing BAL to peripheral blood in TB patients have shown that bronchoalveolar cells from affected lung segments respond more strongly to mycobacterial antigens than PBMCs from the same patient do, and that pulmonary TB is characterized by heightened local Th1 activity (elevated IL-12/IFN-γ) at the site of disease even where the peripheral response is comparatively blunted.

This compartmentalization is the scientific rationale for the project: a BAL-based cytokine panel is modelling a genuinely different (and less thoroughly mined) biomarker space than blood-based IGRA/Luminex panels, while still using the same multiplex-immunoassay data structure and analytical skillset.

The clinical framing follows the WHO's Target Product Profile (TPP) for a TB triage test, which specifies a minimum acceptable performance of ≥90% sensitivity and ≥70% specificity (optimal ≥95%/≥80%). This is a useful, realistic benchmark to report classifier performance against, rather than reporting AUC alone. Notably, IP-10 already features in a real host-protein triage signature (alongside CRP and SAA) developed by the EDCTP Triage Consortium, which is part of why IP-10 anchors the panel simulated here.

2. Simulated Study Design

Groups	Active TB (n=150), LTBI (n=150), Uninfected (n=120)
Sample type	Bronchoalveolar lavage (BAL) supernatant, unstimulated
Cytokine panel	IP-10 (CXCL10), IL-6, IL-1β, IFN-γ, TNF-α, IL-8
Covariates	Age, sex, HIV status, BAL fluid volume
Total n	420

HIV co-infection prevalence was simulated higher in the Active TB group (30%) than in LTBI (15%) or Uninfected (8%) groups, reflecting the TB-HIV syndemic context of South African TB epidemiology rather than a general-population HIV prevalence.

3. Simulation Methodology

3.1 Correlated cytokine generation (two-factor model)

Cytokines were not drawn independently. Each subject was assigned two latent immune-activity scores — a "Th1 axis" (loading onto IP-10, IFN-γ, TNF-α) and an "innate/neutrophil axis" (loading onto IL-6, IL-1β, IL-8) — and each cytokine's log-concentration was built from a group-level mean, an HIV adjustment, a weighted combination of the two latent factors, and cytokine-specific residual noise. This reproduces realistic within-panel correlation structure instead of six unrelated random variables.

3.2 Group and HIV effects

Group-level fold-changes (multiplicative, applied on the log scale relative to the Uninfected baseline) were chosen to reflect the qualitative direction and rough magnitude of effects reported in the BAL/TB immunology literature above — largest elevation in Active TB, intermediate in LTBI, strongest signal in the Th1-axis cytokines (IP-10, IFN-γ, TNF-α). HIV-positive status attenuates the Th1-axis cytokines more than the innate-axis cytokines (IFN-γ ×0.55, TNF-α ×0.65, IP-10 ×0.85; IL-6/IL-1β/IL-8 unaffected), consistent with CD4-dependent Th1 responses being disproportionately compromised by HIV co-infection while innate/neutrophil signalling is comparatively preserved. These multipliers are simulation parameters chosen to be directionally realistic — not measured effect sizes from any single study — and are documented in full in the simulation script.

3.3 Assay noise

The six cytokine columns provided are clean BAL supernatant concentrations — each subject's true underlying value (from the group/HIV/factor model above) with a 12% log-normal coefficient of variation applied, representative of typical multiplex bead-assay (e.g. Luminex) technical variability. No dilution correction is required: the values can be used directly for group comparison, correlation, and classification analyses.

Note: real BAL studies do need to correct for variable saline-dilution of the sample (commonly via a urea-ratio method) before absolute concentrations are biologically interpretable. That correction step is deliberately not included here to keep the dataset usable with standard regression/ML workflows; it's flagged in section 5 as an optional extension if you want to add that layer later.

4. Data Dictionary

Column	Description
subject_id	Unique simulated subject identifier
group	Active_TB / LTBI / Uninfected
age_years	Age in years
sex	Male / Female
hiv_status	Positive / Negative
bal_volume_ml	Simulated volume of BAL fluid recovered (mL) — descriptive covariate
IP10_bal_pg_ml	BAL IP-10 / CXCL10 concentration (pg/mL)
IL6_bal_pg_ml	BAL IL-6 concentration (pg/mL)
IL1B_bal_pg_ml	BAL IL-1β concentration (pg/mL)
IFNG_bal_pg_ml	BAL IFN-γ concentration (pg/mL)
TNFA_bal_pg_ml	BAL TNF-α concentration (pg/mL)
IL8_bal_pg_ml	BAL IL-8 concentration (pg/mL)

5. Suggested Analysis Workflow

1. Exploratory analysis: group comparisons, correlation heatmaps, PCA/UMAP of the cytokine space.
2. Classifier: nested cross-validation comparing logistic regression against random forest/XGBoost for 3-class discrimination (Active TB / LTBI / Uninfected); report multiclass AUC and calibration.
3. Model explanation: SHAP or permutation importance to identify which cytokines drive classification.
4. Clinical framing: report sensitivity/specificity against the WHO TPP triage-test benchmark (≥90%/≥70%), and consider a decision-curve analysis.
5. Optional extensions: HIV-stratified subgroup analysis; a limit-of-detection/censored-data sensitivity analysis (real multiplex assays commonly have some cytokines fall below the assay's detection limit, particularly in lower-signal groups); adding a BAL dilution/urea-correction layer later if you want to take that on as a follow-up methods exercise.

6. References

- "Human Pulmonary Tuberculosis: Understanding the Immune Response in the Bronchoalveolar System." Biomolecules, 2022. https://pmc.ncbi.nlm.nih.gov/articles/PMC9405639/
- "Local Pulmonary Immunological Biomarkers in Tuberculosis." PMC. https://pmc.ncbi.nlm.nih.gov/articles/PMC7973084/

- "Field evaluation of a point-of-care triage test for active tuberculosis (TriageTB)." https://pubmed.ncbi.nlm.nih.gov/37400753/ — describes the CRP/SAA/IP-10 host-protein triage signature and the WHO TPP benchmark of ≥90% sensitivity / ≥70% specificity.
- "Guiding the development of the tuberculosis screening target product profile using single-screen and multi-screen approaches: a modelling study." Lancet Global Health, 2025.

This document and the accompanying dataset were prepared as a methods/portfolio project. Simulation parameters are documented in simulate_bal_cytokines.py for full reproducibility and so that every numeric choice can be audited or adjusted.